A Computational Adaptation of Vernon L. Smith's Experimental Study of Competitive Market Behavior Using Large Language Models

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Research area: Computational Economics, Experimental Economics, and AI-Based Economic Simulation

1. Research Objective

This project develops an open-source computational double-auction market populated by large language model (LLM) agents. Based on Vernon L. Smith's (1962) experimental study of competitive market behavior and its competitive double auction design, it examines whether changes in prompt instructions influence price formation, trading activity, and allocative efficiency.

The implementation uses phi3:mini, a locally deployed open-weights language model, to make agent instructions, market rules, and inference settings available for inspection. Three prompting treatments—Minimal, Standard, and Scaffolded—allow preliminary comparisons of how strategic objectives and structured decision guidance affect market behavior. The project is a proof of concept intended to establish computational feasibility and identify behavioral patterns for further investigation, rather than to establish general causal effects.

2. Economic Framework and Experimental Design

The market follows a continuous double-auction mechanism in which buyers submit bids and sellers submit asks. Agents may accept standing offers or pass on an opportunity to trade. The environment comprises 11 buyers and 11 sellers over five consecutive trading periods, with each active trader permitted to exchange one unit per period. Agents act sequentially, and completed trades remove the participating buyer and seller from the market for the remainder of that period.

Agents receive private information about their roles and reservation values or costs, alongside public information about outstanding bids and asks, transaction prices, and market history. The simulation uses a randomized trading schedule and fixed experimental seeds (valuation seed: 42, scheduler seed: 42). The language model runs locally through Ollama with a temperature setting of 0.7.

The reservation schedules are defined as follows:

$$\text{Buyer Values } (V_i) \in \{0.75, 1.00, 1.25, \ldots, 3.25\}$$
$$\text{Seller Costs } (C_j) \in \{0.75, 1.00, 1.25, \ldots, 3.25\}$$

The experimental benchmark is a competitive equilibrium price of 2.00 and a quantity of six units per period, including the zero-surplus marginal unit. The corresponding maximum attainable total surplus is 7.50 per period (37.50 across five continuous trading periods). These benchmarks follow from the adapted reservation schedules rather than from a claim that the original Smith experiment used identical numerical values.

The experimental design varies the strategic guidance provided in the system prompt while retaining the underlying market rules and valuation schedules:

- **Minimal:** Specifies the agent's role, private reservation value or cost, permitted actions, and basic market rules.
- **Standard:** Adds an explicit instruction to maximize individual trading profit.
- **Scaffolded:** Introduces a structured decision checklist covering reservation values, available offers, transaction history, and previous actions.

Smith's original experiments provide the human-trader reference against which the simulated market's trading outcomes and price behavior can be assessed.

3. Preliminary Results

Each prompting treatment has been evaluated in one exploratory run. The results indicate substantial differences in market performance.

Measure	Minimal	Standard	Scaffolded
Completed trades	20	18	2
Mean transaction price	2.2000	1.5139	0.9500
Total realized surplus	18.75	24.75	2.75
Allocative efficiency	50.00%	66.00%	7.33%
Agent decisions	297	290	323
Runtime (minutes)	78.54	84.00	88.08

The Standard treatment achieved the highest reported allocative efficiency and realized surplus. Explicit profit-maximization instructions may have helped agents pursue individual gains more effectively than the basic instructions alone. The Scaffolded treatment performed poorly, with only two completed trades and substantially lower realized surplus. Its decision logs indicated frequent pass actions and limited market activity, suggesting that additional procedural guidance may hinder decision-making in a small language model.

None of the treatments reproduced the strong convergence toward competitive outcomes associated with Smith's human-subject experiments. The results instead demonstrate that simulated market behavior may be highly sensitive to the strategic guidance provided in agent prompts. Changing the system prompt can alter the agent's risk tolerance and decision-making logic, meaning the prompt effectively functions as a market rule. These initial results suggest that evaluating algorithmic price discovery requires analyzing prompt design alongside traditional auction rules.

4. Methodological Contributions and Limitations

The project provides a locally executable market simulation in which agent instructions and trading rules can be inspected and modified. Using an open-weights model through Ollama reduces dependence on proprietary inference services and allows the experimental setup to be documented. However, local deployment alone does not guarantee identical outputs across executions or software environments. Reproducibility also depends on model versions, inference settings, randomization, and implementation details.

The principal limitations are the small model, the single-run design, and the absence of comprehensive machine-readable records of every decision and quotation. The preliminary findings therefore serve primarily to identify implementation issues and motivate more systematic experiments.

5. Research Directions

Further work will extend the project in four directions.

- Repeated runs across independent random seeds will help estimate variation in transaction prices, realized surplus, allocative efficiency, and convergence.
- Comparisons across larger open-weights models will examine whether model capacity changes the effects of structured prompting.
- Alternative supply-and-demand schedules will allow investigation of price adjustment under asymmetric trading incentives, including the excess-rent hypothesis discussed by Smith.
- Decision-level logging will support a closer analysis of bidding behavior, rejected offers, pass actions, and changes in trading strategies across periods.

The broader objective is to establish a transparent computational framework for studying how language-model agents behave within established economic institutions. The current implementation provides an initial basis for that research, while its findings remain exploratory.